PAPER ID-311382

Subject Code: BCE055

Roll No: | | | | | | | | | | | | | |

BTECH
(SEM V) THEORY EXAMINATION 2025-26
ENGINEERING HYDROLOGY

TIME: 3 HRS **M.MARKS: 70**

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A

1. Attempt *all* questions in brief. **2 x 07 = 14**

Q no.	Question	CO	Level
a.	What is a water budget equation?	CO1	K1
b.	Define precipitation.	CO1	K1
c.	Name any three factors affecting runoff from a catchment.	CO2	K1
d.	What is the basic assumption of unit hydrograph theory?	CO2	K1
e.	What is a runoff coefficient in the rational method?	CO3	K1
f.	What is an unconfined aquifer?	CO4	K1
g.	What is meant by control of groundwater contamination?	CO5	K1

SECTION B

2. Attempt any *three* of the following: **07 x 3 = 21**

a.	Explain the importance of water balance studies in hydrology and water resources engineering.	CO1	K2
b.	Define a synthetic unit hydrograph. Also explain the need for synthetic unit hydrograph.	CO2	K2
c.	Discuss hydrologic and hydraulic flood routing, highlighting their applications.	CO3	K2
d.	Explain forms of subsurface water, aquifers, and groundwater occurrence.	CO4	K2
e.	Explain the design and working of rooftop rainwater harvesting systems.	CO5	K2

SECTION C

3. Attempt any *one* part of the following: **07 x 1 = 07**

a.	Consider station A is located at the origin of a rectangular coordinate system. Four surrounding rain-gauge stations nearest to A are located at coordinates (20, 30) km, (−10, 15) km, (−15, −10) km, and (10, −5) km. During a storm, Station A failed to record rainfall. The recorded storm rainfalls at the four surrounding stations are 75 mm, 90 mm, 70 mm, and 60 mm, respectively. Estimate the missing rainfall at Station A.	CO1	K3
b.	Explain the factors affecting evaporation from water surfaces. Alos explain different methods of estimation of evaporation.	CO1	K2

4. Attempt any *one* part of the following: **07 x 1 = 07**

a.	Explain the runoff characteristics of a stream and factors affecting runoff.	CO2	K2
b.	Define a hydrograph and explain its different components with a neat sketch.	CO2	K2

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5. Attempt any *one* part of the following: 07 x 1 = 07

a.	Explain the Rational Method for flood estimation. State its assumptions, merits, and limitations.	CO3	K2
b.	Explain the role of reservoirs, levees, and channel improvement in flood control.	CO3	K2

6. Attempt any *one* part of the following: 07 x 1 = 07

a.	Explain unsteady flow through confined aquifers and its practical importance.	CO4	K2
b.	Define specific capacity of a well and explain its significance.	CO4	K2

7. Attempt any *one* part of the following: 07 x 1 = 07

a.	Explain the steps involved in well construction, including drilling methods.	CO5	K2
b.	Explain methods for preventing clogging and corrosion in wells.	CO5	K2

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